

HYBRID INTELLIGENCES

Embodied Leadership and Creativity in the Era of AI

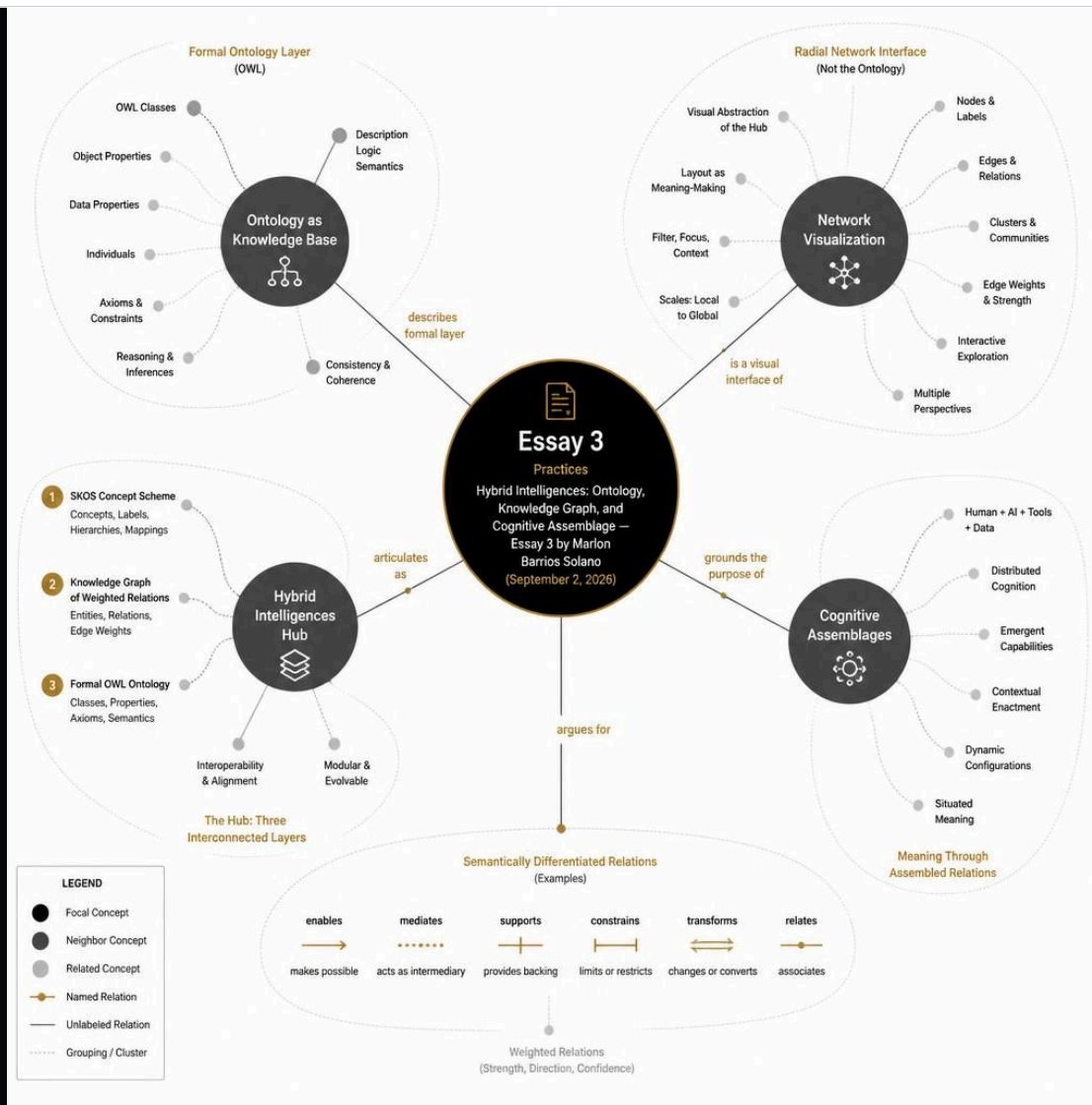
A hybrid dynamic knowledge architecture of concepts, essays, visualization, conversational AI, documentation, and program materials for embodied leadership and creativity in the era of AI.

ESSAY 3

Hybrid Intelligences: Ontology, Knowledge Graph, and Cognitive Assemblage

Marlon Barrios Solano · September 2, 2026

PROGRAM	Hybrid Intelligences · Creative B Summer · July 13–30, 2026
INSTITUTION	University of Florida · College of the Arts (COA)
PARTNERS	Center for Arts, Migration + Entrepreneurship (CAME) · Center for Arts in Medicine (CAM) · IGNITE Engineering · Wertheim Laboratory
CO-DIRECTORS	Marlon Barrios Solano · Erika Moore
CONCEPTUAL NETWORK, DEVELOPMENT & ONTOLOGY	Marlon Barrios Solano
DOCUMENT TYPE	Program essay · PDF export from the Hybrid Intelligences site
RELATED MATERIALS	Conceptual network · Ontology (JSON-LD / Turtle / OWL) · Conversational AI (Voice) · Creative B 2026 (Canvas, slides, highlights, lobby showcase)



Essay 3 as practice · four clusters and a vocabulary of relations

In Hybrid Intelligences, I am developing an evolving knowledge architecture that combines a computational ontology, a knowledge graph, artistic research, network visualization, and generative AI interfaces. Together, these components form an experimental cognitive assemblage through which I investigate intelligence as coupling across bodies, technologies, institutions, environments, and worlds.

I use the term ontology deliberately. In knowledge representation and artificial intelligence, an ontology is not simply a collection of keywords, concepts, or visualizations. It provides a formal and machine-readable way of specifying entities and concepts within a domain, their categories, properties, and relationships.

The Hybrid Intelligences Hub currently contains 238 concepts and 2,379 weighted relationships. One hundred two of those relationships are also typed: directed verbs such as couples with, enables, mediates, and proposes, published in the network and in the machine-readable ontology. The concepts and relationships are not only displayed through a visualization; I also represent them through Semantic Web and knowledge-representation standards, including SKOS, JSON-LD, RDF/Turtle, and OWL.

The project publishes several machine-readable representations of this knowledge architecture:

```
ontology.jsonld  ontology.ttl  ontology.owl.ttl
```

Within the OWL representation, I define `hi:Concept` as a superclass and organize concepts through classes such as:

```
hi:ProgramConcept  hi:OrganizationConcept  hi:PremiseConcept  hi:ParticipantConcept  
hi:BackgroundConcept  hi:FacilitatorConcept  hi:PracticeConcept  hi:TensionConcept  
hi:QualityConcept  hi:PhenomenonConcept  hi:DomainConcept  hi:FrameworkConcept  
hi:AuthorConcept
```

The OWL layer also defines object properties. `hi:relatedTo` remains the general, weighted association. Thirteen directed verbs are declared as `rdfs:subPropertyOf hi:relatedTo`:

```
hi:couplesWith  hi:enables  hi:mediates  hi:cultivates  hi:constrains  
hi:participatesIn  hi:critiques  hi:proposes  hi:instantiates  hi:develops  
hi:enacts  hi:embodies  hi:emergesFrom
```

I also define datatype properties, domains, ranges, subclasses, and individuals. In this sense, the project goes beyond producing a visualization of interconnected ideas. I am

engaging directly with practices of ontology engineering and computational knowledge representation.

THREE INTERCONNECTED LAYERS

I understand the current architecture of Hybrid Intelligences as operating across three interconnected layers: a concept scheme or knowledge organization system, a knowledge graph, and a formal ontology.

At the level of knowledge organization, I use SKOS structures such as `skos:ConceptScheme`, `skos:prefLabel`, `skos:definition`, `skos:broader`, `skos:inScheme`, and `skos:topConceptOf`. These allow me to define concepts, descriptions, categories, and conceptual hierarchies in a structured and machine-readable form.

At the level of the knowledge graph, I instantiate hundreds of concepts and connect them through thousands of relationships. The interactive radial network is therefore not the ontology itself. It is one visual and navigational interface into the underlying knowledge architecture.

The OWL layer extends this further by formally defining classes, subclasses, object properties, datatype properties, domains, ranges, and individuals. This is where Hybrid Intelligences becomes an ontology in the stronger sense associated with Semantic Web technologies and computational knowledge representation.

I therefore understand the architecture approximately as:

Ontology → Knowledge Graph → Interfaces

But the actual research process is recursive rather than linear:

Essays + artistic research → Ontology → Knowledge Graph → Network / Voice / Image
/ Mini-pod / Enact → New research → Ontology

The system continually feeds back into itself.

FROM ASSOCIATIONS TO SEMANTIC RELATIONSHIPS

Many of the relationships still operate through a general formulation of relatedness plus relational weight. That substrate remains: thousands of `relatedTo` edges whose thickness is strength. It has been useful for constructing and navigating the network. I have now also begun to ask not only whether two concepts are connected, but *how* they are connected.

The general association is still:

- Embodiment → relatedTo → Cognition

On top of it the graph now carries a seed of semantically differentiated, directed relationships — one hundred two assertions using the thirteen verbs named above. They are visible in the network (a Relations filter, combinable with a category ring), in the ontology browser, and as OWL object-property assertions such as `hi:body` `hi:couplesWith` `hi:ai`.

Examples now asserted in the Hub:

- body → couples with → artificial intelligence
- embodiment → enables → cognition
- LLM → mediates → cognition
- somatics → cultivates → literacies
- CAME → constrains → creativity
- conversational AI → participates in → Hybrid Intelligences Hub
- hybrid intelligences → critiques → techno-dualism
- Hayles → proposes → cognitive assemblages
- Maturana and Varela → propose → autopoiesis

- Marlon Barrios Solano → proposes → Essay 1, Essay 2, Essay 3
- network visualization → instantiates → assemblage
- Essay 3 → develops → ontology as knowledge base
- Enact → enacts → coupling
- body → embodies → cognition
- intelligence → emerges from → coupling

Most edges remain untyped association. The seed is small on purpose. I am not pretending that every proximity is already a differentiated verb.

This shifts the research question from:

What concepts are connected?

toward:

What kinds of relationships produce the assemblage?

That question is no longer only prospective. It is now a working layer of the knowledge architecture — still thin, still growing, already inspectable.

QUESTIONING FIXED CATEGORIES

This development also raises a productive problem within the ontology itself.

Some of the current top-level categories are represented as mutually disjoint classes. I am interested in reconsidering this structure because the theoretical proposition of Hybrid Intelligences challenges precisely this kind of rigid separation.

A practice might simultaneously function as a framework. An artwork might be a practice, an experiment, a conceptual model, and part of a program. A person might simultaneously operate as an author, participant, facilitator, artist, and researcher.

Rather than seeing this ambiguity as something that needs to be eliminated from the ontology, I am interested in treating it as a research problem.

This leads to a central question:

What would an ontology of cognitive assemblages look like if entities were allowed to occupy multiple functional positions simultaneously?

For me, this question connects the technical construction of the ontology directly with the philosophical and artistic research of Hybrid Intelligences.

ONTOLOGY AND GENERATIVE AI

The ontology also performs a specific function in relation to generative AI.

I do not understand it as training or fine-tuning a large language model unless model weights are actually being modified. Instead, the ontology provides a structured knowledge and epistemic context for the language model.

The conversational AI within the Hub is therefore grounded in the Hybrid Intelligences ontology and research framework.

This distinction is important because an ontology and a large language model organize knowledge differently.

An LLM operates through distributed statistical representations learned from very large quantities of language. The ontology allows me to make particular conceptual and epistemic commitments explicit:

- These are the entities I consider relevant.
- These are their definitions.
- These are the categories I am proposing.
- These are the relationships I consider meaningful.

The language model contributes something different: linguistic flexibility, interpretation, conversation, recombination, and generativity.

The resulting assemblage can be understood as:

formal and semi-formal knowledge structures + probabilistic language models +
human interaction + embodied practices + artistic research

This is where the technical architecture of the project intersects with its larger conceptual proposition.

I am not proposing that the AI itself is the intelligence.

The ontology is not the intelligence.

The human is not the intelligence.

The network is not the intelligence.

Intelligence emerges through their coupling.

This is the central proposition I am testing through Hybrid Intelligences.

THE HUB AS AN EXPERIMENTAL COGNITIVE ASSEMBLAGE

I therefore describe Hybrid Intelligences as an evolving knowledge architecture combining a computational ontology, knowledge graph, artistic research, network visualization, and generative AI interfaces. Together, these components form an experimental cognitive assemblage for investigating intelligence as coupling across bodies, tools, technologies, institutions, environments, and worlds.

The different components of the Hub—Ontology, Network, Voice, Image, Mini-pod, Enact, essays, and artistic works—are not separate tools. They are different modes for

entering and activating the same evolving knowledge ecology.

Writing feeds the ontology.

The ontology reorganizes the network.

The network and ontology ground conversations.

Concepts become images, voices, audio, and embodied prompts.

Artistic practice enters the assemblage and generates new questions.

Those questions return to the research and potentially transform the ontology.

For this reason, I see the next stage of Hybrid Intelligences not simply as expanding the number of concepts or adding new AI features. Relations such as couples with, enables, mediates, cultivates, constrains, participates in, critiques, proposes, instantiates, develops, enacts, embodies, and emerges from are now first-class components of the knowledge architecture. What remains is to grow that seed, to let more of the graph speak in verbs, and to keep asking whether entities may occupy more than one functional position at once.

This makes the ontology itself part of my artistic and epistemological research.

Hybrid Intelligences is therefore not simply a project about hybrid intelligence. The architecture, the AI systems, the ontology, my artistic practice, its users, and the institutions within which it develops participate in producing the cognitive assemblage that the project is attempting to understand.

Perhaps the Hub is not simply about Hybrid Intelligences.

It is an experiment in becoming one.

REFERENCES

- Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Margaret Mitchell. “On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?” In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. 2021.
- Berners-Lee, Tim, James Hendler, and Ora Lassila. “The Semantic Web.” *Scientific American* 284, no. 5 (2001): 34–43.
- Clark, Andy. *Supersizing the Mind: Embodiment, Action, and Cognitive Extension*. Oxford: Oxford University Press, 2008.
- Clark, Andy, and David J. Chalmers. “The Extended Mind.” *Analysis* 58, no. 1 (1998): 7–19.
- Deleuze, Gilles, and Félix Guattari. *A Thousand Plateaus: Capitalism and Schizophrenia*. Translated by Brian Massumi. Minneapolis: University of Minnesota Press, 1987.
- Gruber, Thomas R. “A Translation Approach to Portable Ontology Specifications.” *Knowledge Acquisition* 5, no. 2 (1993): 199–220.
- Guarino, Nicola, Daniel Oberle, and Steffen Staab. “What Is an Ontology?” In *Handbook on Ontologies*, edited by Steffen Staab and Rudi Studer, 1–17. Berlin: Springer, 2009.
- Hayles, N. Katherine. *Bacteria to AI: Human Futures with Our Nonhuman Symbionts*. Chicago: University of Chicago Press, 2025.
- Hayles, N. Katherine. *How We Became Posthuman: Virtual Bodies in Cybernetics, Literature, and Informatics*. Chicago: University of Chicago Press, 1999.
- Hayles, N. Katherine. *Unthought: The Power of the Cognitive Nonconscious*. Chicago: University of Chicago Press, 2017.
- Hogan, Aidan, et al. “Knowledge Graphs.” *ACM Computing Surveys* 54, no. 4 (2021): Article 71.
- Hutchins, Edwin. *Cognition in the Wild*. Cambridge, MA: MIT Press, 1995.
- Lewis, Patrick, et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” *Advances in Neural Information Processing Systems* 33 (2020): 9459–9474.
- Noy, Natalya F., and Deborah L. McGuinness. *Ontology Development 101: A Guide to Creating Your First Ontology*. Stanford Knowledge Systems Laboratory Technical Report KSL-01-05, 2001.

Simondon, Gilbert. *On the Mode of Existence of Technical Objects*. Translated by Cecile Malaspina and John Rogove. Minneapolis: Univocal Publishing, 2017.

World Wide Web Consortium (W3C). *OWL 2 Web Ontology Language Document Overview*. W3C Recommendation, December 11, 2012.

World Wide Web Consortium (W3C). *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, August 18, 2009.